The Epistemic Mirror

Goal Attribution and the Relational Nature of AI Understanding

Soumya Banerjee

Computer Science, University of York, York, UK

The central question

- The challenge is not only to define what a system does, but to achieve an "understanding of understanding"
- Large Language Models present a wall of complexity that makes mechanical explanations of little use for daily interaction
- Is the "understanding" we attribute to these systems:
  - a property residing in their silicon architecture, or
  - a psychological projection emerging from the human observer?

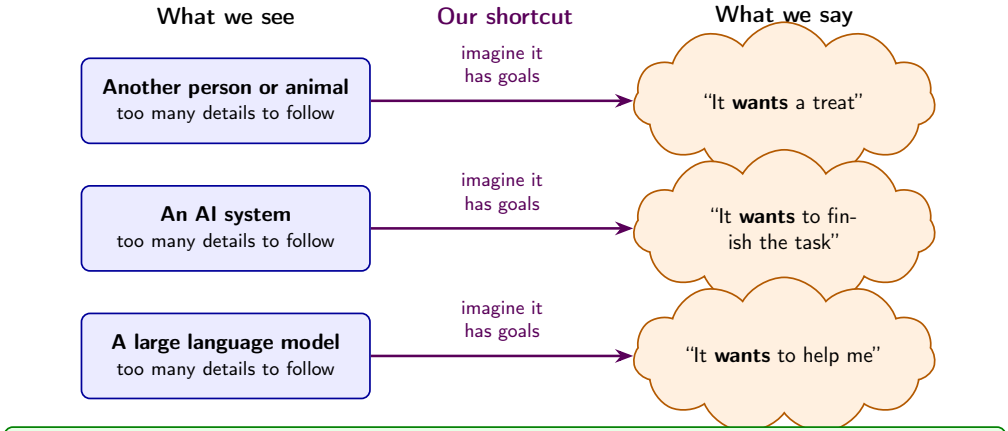
Thesis

Teleological projection

Attributing goals to a system is not a failure of scientific objectivity. It is a sophisticated and necessary epistemological strategy.

- Treating a statistical model as if it has intentions, desires, or goals translates a high-dimensional mathematical reality into a low-dimensional narrative structure
- The shift is from the "ontic" truth of what a system is to an "epistemic" unity between the observer's mental frameworks and the system's observed behaviours

The idea in one picture



Goals are a shortcut for understanding.
They help us explain and predict what a system will do, without tracking every detail inside it. They are a useful lens, not a claim that the system has an inner life.

What does “understanding” mean?

- In traditional semantics, understanding means correctly identifying the rules and concepts behind a statement
- Directed at reality itself, the “phenomenal world,” understanding becomes far more elusive
- We do not understand a human peer by calculating the firing of every neuron in their brain
  - The task is computationally impossible
  - It provides no social utility

The intentional stance

- Dennett (1987) calls the alternative the "intentional stance"
- We assume the other person is a rational agent with goals and beliefs
- This assumption lets us predict their behaviour with considerable accuracy

Central claim

We are now being forced to adopt the same stance toward artificial systems.

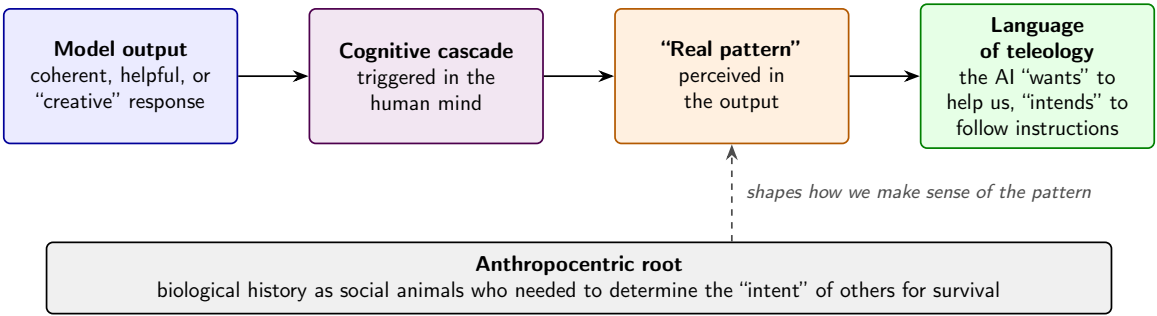
Why adopt it for AI?

- Not because these systems have “souls” or “internal lives” similar to our own
- Because their complexity has reached a threshold where any other form of understanding fails to provide predictive or interactive power

Projection is an anthropocentric act

- Rooted in our biological history as social animals who needed to determine the “intent” of others for survival
- A coherent, helpful, or “creative” model response triggers a cognitive cascade in the human mind
- We perceive a “real pattern” in the output and wrap it in the language of teleology
- We say the AI “wants” to help us or “intends” to follow our instructions

From model output to attributed goal



Understanding as a relational event

- The "understanding of understanding" recognises that our own cognitive architecture actively participates in the creation of meaning
- We are not passive mirrors reflecting a mechanical reality; we are active projectors who impose order on complexity through the lens of agency

Relational view

Understanding is not a one-way street from subject to object. It occurs at the interface between the system's structural constraints and the observer's interpretative needs.

Two stances, one tension

Physical stance

- The world of silicon and mathematics

Intentional stance

- The world of goals and meaning

- Kant (1790) suggested that viewing things as if they had a purpose is a subjective necessity when we encounter the organised complexity of nature
- We cannot help but see purpose in a bird's wing or the growth of a tree, even if the underlying physics is blind forces

Regulatory, not intrinsic, teleology

- We cannot help but see purpose in a machine that mimics our language perfectly
- This does not mean the machine has an "intrinsic" goal, in the way an acorn "wants" to become an oak tree
- It suggests a "regulatory" teleology: the goal is a tool the observer uses to render the system intelligible
- The "understanding" is the successful navigation of the system's behaviour using these projected categories

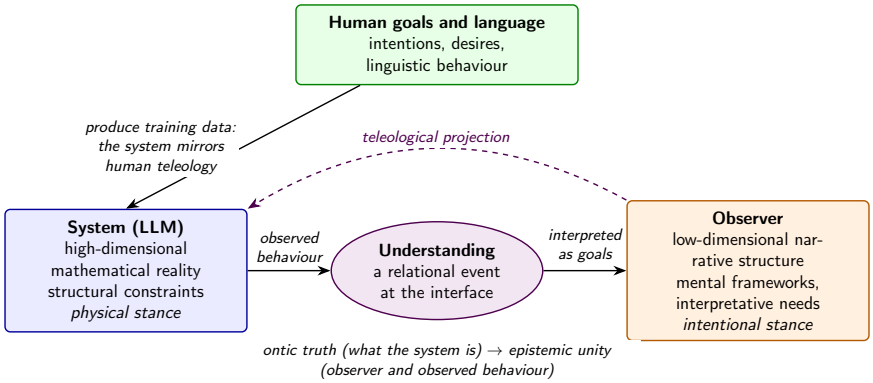
Ontic and epistemic unity

- There is no clean separation between how a system works and how we understand it
- The "meaning" of an AI's output is not just in the bits and bytes, nor just in the user's head
- It exists in the successful interaction between the two

The epistemic mirror

- Attributing a goal to an LLM identifies a "real pattern" in data produced by humans who themselves have goals
- The AI becomes a mirror of human teleology, so understanding it is a form of self-reflection

Summary figure



Elusiveness and the middle ground

- Understanding is elusive because it is dynamic, shifting with the level of abstraction
- Too close: the "goal" disappears into a sea of numbers
- Too far: mysticism that treats the machine as a conscious deity
- The middle ground: use teleological projection as an epistemic tool while remaining aware that it is a projection
- Our anthropocentric bias is not a flaw to be removed but the foundation of our ability to make sense of a complex and automated world

Conclusions and references

- Understanding is not the discovery of a hidden essence in a target system, but the construction of a bridge between its output and our mental models
- Goal attribution is an observer-relative strategy that reconciles the mechanical reality of the machine with the semantic reality of human life
- We must navigate the phenomenal world as participants in a shared, projected reality of intent and purpose

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